Demographic Predictors of Reform UK Vote Share in the 2024 General Election

Anton Robb

Contents

1 Introduction **1**

2 Preparing the data **2**

3 Exploring the response **2**

4 Univariate associations **3**

5 Regional variation **4**

6 A combined ranking of marginal associations **5**

7 Multiple regression **5**

7.1 Model diagnostics 6

7.2 Model summary statistics 7

8 Univariate versus multivariate: where the story changes **8**

9 Interpretation **9**

10 Notes **10**

1 Introduction

The 2024 UK general election saw Reform UK emerge as a significant electoral force, with support varying widely across the country's 650 parliamentary constituencies. This project asks a simple question: which demographic and socioeconomic characteristics of a constituency best predict how strongly it voted for Reform UK?

To answer it, I combine constituency-level election results with 2021 census demographics (age structure, ethnic composition, educational attainment, benefit-claimant rates and median wages) and constituency geography. I build a multiple linear regression model of Reform's vote share, and pay particular attention to a statistical subtlety that turns out to matter a great deal here: how the apparent effect of a variable can change, and even reverse, once other correlated variables are accounted for.

All data are public, sourced from the House of Commons Library and the 2021 census. Data preparation, modelling and evaluation are carried out entirely in R.

2 Preparing the data

The analysis draws on several separate files, one per data domain, each keyed by a constituency identifier code. I load each, keep a focused set of columns to avoid redundancy and overfitting, and merge them into a single constituency-level frame.

```
library(dplyr)

elec  <- read.csv("election2024.csv")
age   <- read.csv("age.csv")
claim <- read.csv("claimants.csv")
eth   <- read.csv("ethnicity.csv")
qual  <- read.csv("qualifications.csv")
wage  <- read.csv("wages.csv")

# Retain a small, non-redundant set of explanatory variables per domain
age   <- age[,   c("ONSConstID", "pct18.24", "pct65.up")]
claim <- claim[, c("ONSConstID", "claimant_rate")]
eth   <- eth[,   c("ONSConstID", "white", "asian", "black")]
qual  <- qual[,  c("ONSConstID", "Qual4_pct", "Qual5_pct")]
wage  <- wage[,  c("ONSConstID", "MedianWage")]

elec$Region <- factor(elec$RegionName)

df <- elec |>
  merge(age,   by = "ONSConstID") |>
  merge(claim, by = "ONSConstID") |>
  merge(eth,   by = "ONSConstID") |>
  merge(qual,  by = "ONSConstID") |>
  merge(wage,  by = "ONSConstID")
```

The response variable is Reform's share of the valid vote in each constituency. I restrict the analysis to constituencies where Reform actually stood a candidate, since a zero vote elsewhere reflects absence rather than lack of support.

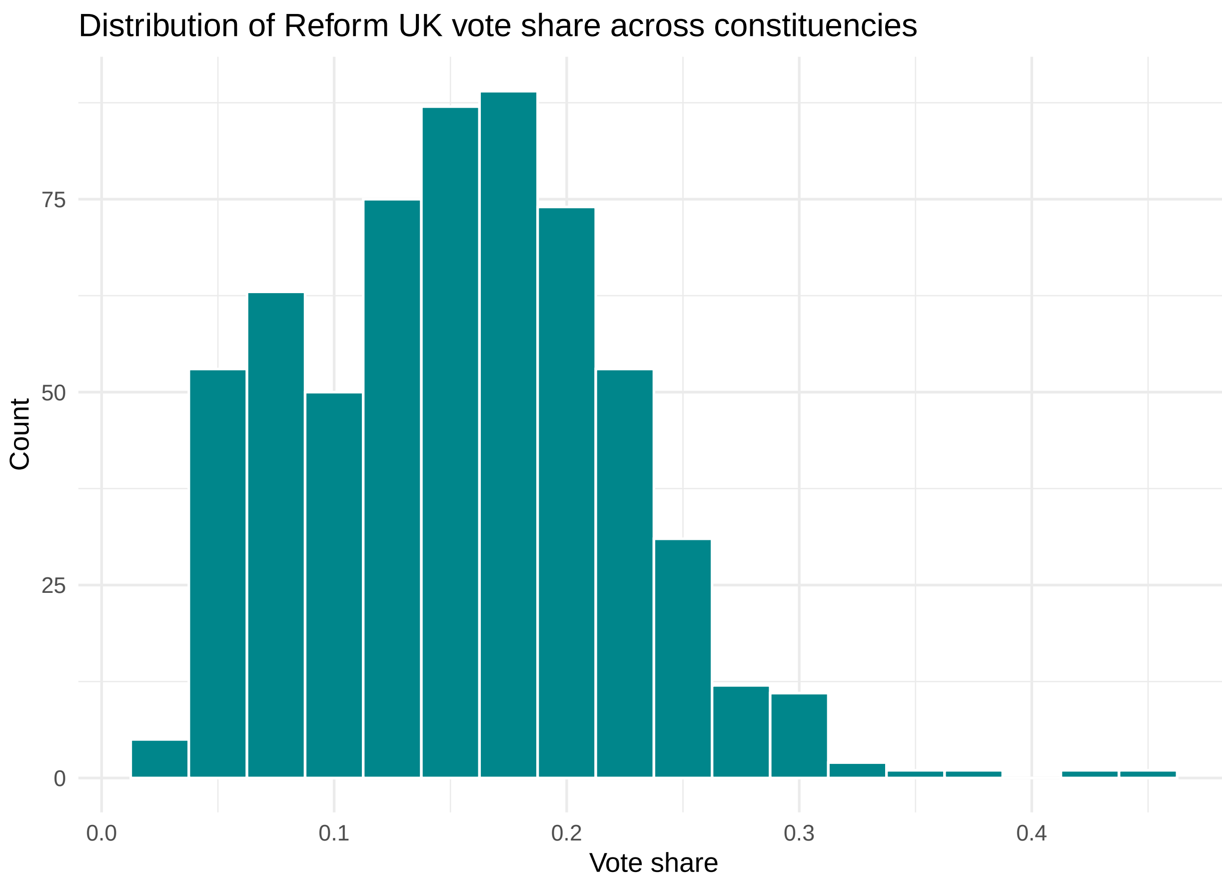
```
df$vote_share <- df$RUK / df$Valid.votes
df <- df[df$RUK > 0, ]
```

I selected the explanatory variables to capture distinct demographic and socioeconomic dimensions while avoiding near-duplicates: two age bands (younger and older adults), three broad ethnic groups, two educational-attainment measures (at opposite ends of the qualification scale), the benefit-claimant rate, median wage, and Region as a categorical geographic factor. The aim was breadth of genuinely different signals rather than many collinear measures of the same underlying trait.

3 Exploring the response

```r
library(ggplot2)

ggplot(df, aes(vote_share)) +
  geom_histogram(binwidth = 0.025, fill = "turquoise4", colour = "white") +
  labs(title = "Distribution of Reform UK vote share across constituencies",
       x = "Vote share", y = "Count") +
  theme_minimal()
```




```r
summary(df$vote_share)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.03085 0.10354 0.15470 0.15403 0.19866 0.46183
```

Reform's vote share is unimodal and mildly right-skewed, concentrated between roughly 5% and 25% per constituency, with a small number of constituencies where Reform performed exceptionally strongly forming a right tail.

4 Univariate associations

As a first pass, I examine each continuous variable's association with vote share individually: the slope of a simple linear fit, the Pearson correlation, and its p-value. This gives a marginal picture, before any variables are considered together.

```r
cont_vars <- c("pct18.24", "pct65.up", "white", "asian", "black",
               "Qual4_pct", "Qual5_pct", "claimant_rate", "MedianWage")

uni <- lapply(cont_vars, function(v) {
  ct <- cor.test(df$vote_share, df[[v]])
  data.frame(variable = v,
             slope = coef(lm(df$vote_share ~ df[[v]]))[2],
             correlation = unname(ct$estimate),
             p_value = ct$p.value)
}) |> do.call(what = rbind)

uni[order(uni$p_value), ]
```

```
##                variable         slope correlation      p_value
## df[[v]]6     Qual5_pct -0.5132312391 -0.68818456 1.234657e-86
## df[[v]]8    MedianWage -0.0003425659 -0.46155625 1.844825e-33
## df[[v]]2         white  0.1596706663  0.40807888 7.696411e-26
## df[[v]]5     Qual4_pct  0.8346793179  0.35617780 1.181098e-19
## df[[v]]4         black -0.4257535388 -0.33913388 7.413062e-18
## df[[v]]1      pct65.up  0.4129079820  0.33393523 2.492515e-17
## df[[v]]3         asian -0.2029476281 -0.32391892 2.417730e-16
## df[[v]]       pct18.24 -0.4967153230 -0.27939292 2.203437e-12
## df[[v]]7 claimant_rate -0.1791321419 -0.05327764 1.891786e-01
```

The strongest single association is with the proportion of residents holding no formal qualifications, which is strongly *negatively* correlated with Reform support. Several other variables (age structure, ethnic composition, median wage) also show clear marginal associations.

5 Regional variation

Region is a categorical variable, so I test its association with vote share using a one-way analysis of variance, and inspect the regional means.

```r
aov_region <- aov(vote_share ~ Region, data = df)
summary(aov_region)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)    
## Region       10  1.122 0.11216   42.49 <2e-16 ***
## Residuals   598  1.579 0.00264                   
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
tapply(df$vote_share, df$Region, mean) |> sort()
```

```
##                 Scotland                   London               South West 
##               0.07005710               0.08688723               0.14919681 
##               South East               North West                    Wales 
##               0.15022616               0.16883167               0.17485949 
##          East of England            West Midlands            East Midlands 
##               0.18264915               0.18614822               0.19066161 
## Yorkshire and The Humber               North East 
##               0.19328149               0.21984683
```

Region is very strongly associated with Reform support: average vote share differs substantially across the country, with the highest support in parts of the North and Midlands and the lowest in London and Scotland. Geography clearly carries real signal.

6 A combined ranking of marginal associations

Bringing the continuous and categorical results together, and correcting the p-values for multiple comparisons, gives a ranked view of which variables are most strongly associated with vote share when each is considered on its own.

```
p_region <- summary(aov_region)[[1]]["Region", "Pr(>F)"]

ranking <- rbind(
  data.frame(variable = cont_vars, p_value = uni$p_value),
  data.frame(variable = "Region", p_value = p_region)
)
ranking$p_adj <- p.adjust(ranking$p_value, method = "bonferroni")
ranking[order(ranking$p_adj), ]
```

```
##         variable      p_value        p_adj
## 7      Qual5_pct 1.234657e-86 1.234657e-85
## 10        Region 2.046309e-63 2.046309e-62
## 9     MedianWage 1.844825e-33 1.844825e-32
## 3          white 7.696411e-26 7.696411e-25
## 6      Qual4_pct 1.181098e-19 1.181098e-18
## 5          black 7.413062e-18 7.413062e-17
## 2       pct65.up 2.492515e-17 2.492515e-16
## 4          asian 2.417730e-16 2.417730e-15
## 1       pct18.24 2.203437e-12 2.203437e-11
## 8  claimant_rate 1.891786e-01 1.000000e+00
```

7 Multiple regression

The univariate view is only a starting point, because the explanatory variables are themselves correlated with one another. To isolate each variable's *independent* contribution, I fit a multiple linear regression including all explanatory variables simultaneously.

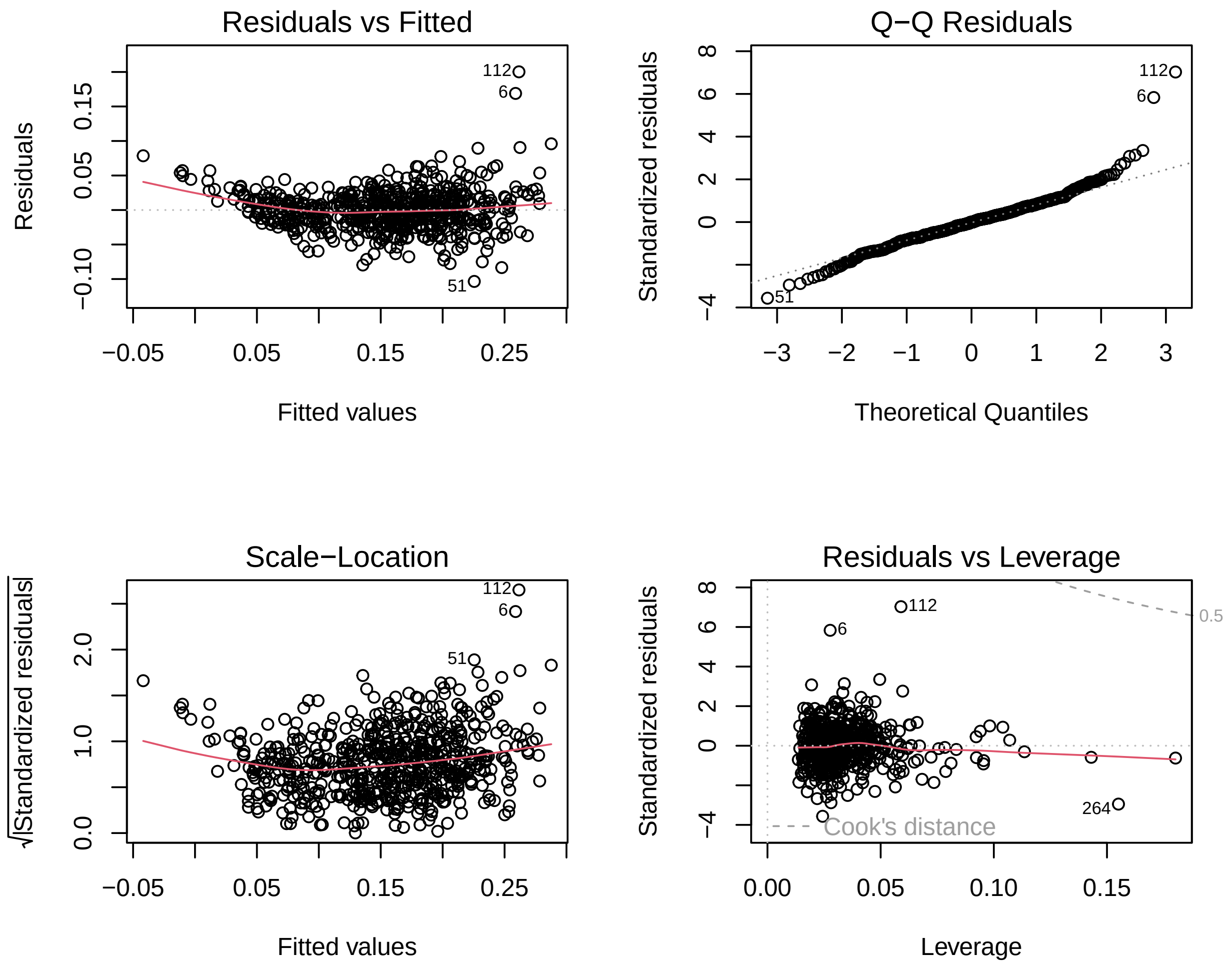
```
multi <- lm(vote_share ~ pct18.24 + pct65.up + white + asian + black +
              Qual4_pct + Qual5_pct + claimant_rate + MedianWage + Region,
            data = df)
summary(multi)
```

```
##
## Call:
## lm(formula = vote_share ~ pct18.24 + pct65.up + white + asian +
##     black + Qual4_pct + Qual5_pct + claimant_rate + MedianWage +
##     Region, data = df)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
```

```
## -0.103467 -0.016816 -0.000226  0.015363  0.200199
##
## Coefficients:
##                                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)                    2.383e-01  9.982e-02   2.387  0.01729 *
## pct18.24                       4.201e-01  7.474e-02   5.620 2.95e-08 ***
## pct65.up                      -2.077e-01  4.404e-02  -4.717 2.99e-06 ***
## white                          3.112e-01  9.457e-02   3.290  0.00106 **
## asian                          8.543e-02  9.574e-02   0.892  0.37256
## black                          1.497e-01  1.147e-01   1.305  0.19253
## Qual4_pct                     -8.790e-01  1.023e-01  -8.595  < 2e-16 ***
## Qual5_pct                     -6.476e-01  3.190e-02 -20.304  < 2e-16 ***
## claimant_rate                 -4.888e-01  1.541e-01  -3.171  0.00160 **
## MedianWage                     6.782e-05  2.737e-05   2.478  0.01350 *
## RegionEast of England         -1.078e-02  6.073e-03  -1.775  0.07633 .
## RegionLondon                   1.747e-02  8.410e-03   2.077  0.03820 *
## RegionNorth East               1.598e-02  7.623e-03   2.096  0.03653 *
## RegionNorth West              -1.168e-02  5.772e-03  -2.023  0.04355 *
## RegionScotland                -1.644e-01  8.069e-03 -20.378  < 2e-16 ***
## RegionSouth East              -1.299e-02  5.620e-03  -2.311  0.02120 *
## RegionSouth West              -2.880e-02  5.985e-03  -4.813 1.89e-06 ***
## RegionWales                   -1.934e-02  7.083e-03  -2.730  0.00652 **
## RegionWest Midlands            1.200e-02  6.075e-03   1.975  0.04870 *
## RegionYorkshire and The Humber -9.005e-04  6.232e-03  -0.144  0.88516
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.02937 on 589 degrees of freedom
## Multiple R-squared:  0.8118, Adjusted R-squared:  0.8057
## F-statistic: 133.7 on 19 and 589 DF,  p-value: < 2.2e-16
```

7.1 Model diagnostics

```r
par(mfrow = c(2, 2))
plot(multi)
```




```r
par(mfrow = c(1, 1))
```

The diagnostic plots show broadly well-behaved residuals: approximately normal with only mild departures in the tails, and roughly constant variance. A small number of high-leverage constituencies are present but none dominate the fit. (Any single constituency with both extreme residual and high leverage would be worth excluding and refitting; the diagnostics here do not indicate a case severe enough to distort the conclusions.)

7.2 Model summary statistics

```r
s <- summary(multi)
e <- residuals(multi); h <- hatvalues(multi)
data.frame(
  df_model      = length(coef(multi)) - 1,
  df_error      = df.residual(multi),
  R_squared     = s$r.squared,
  adj_R_squared = s$adj.r.squared,
```

```
  residual_SE   = s$sigma,
  AIC           = AIC(multi),
  PRESS         = sum((e / (1 - h))^2)
)
```

```
##   df_model df_error R_squared adj_R_squared residual_SE       AIC     PRESS
## 1       19      589 0.8117767     0.8057049  0.02937488 -2546.704 0.5476008
```

The model explains a large majority of the variation in Reform vote share across constituencies, indicating that constituency demographics account for much of the geographic pattern in support.

8 Univariate versus multivariate: where the story changes

The most interesting statistical feature of this analysis is how variables behave differently in isolation versus in the full model. Comparing each variable's covariate-adjusted p-value from the multiple regression against its marginal p-value reveals three distinct patterns.

```
coef_tab <- s$coefficients
p_cont_multi <- coef_tab[cont_vars, "Pr(>|t|)"]
p_region_multi <- anova(multi)["Region", "Pr(>F)"]

compare <- data.frame(
  variable = c(cont_vars, "Region"),
  p_univariate = ranking$p_value[match(c(cont_vars, "Region"), ranking$variable)],
  p_multivariate = c(p_cont_multi, p_region_multi)
)
compare[order(compare$p_multivariate), ]
```

```
##                     variable p_univariate p_multivariate
##                       Region 2.046309e-63   1.956494e-91
## Qual5_pct          Qual5_pct 1.234657e-86   7.028725e-70
## Qual4_pct          Qual4_pct 1.181098e-19   7.503903e-17
## pct18.24            pct18.24 2.203437e-12   2.946627e-08
## pct65.up            pct65.up 2.492515e-17   2.993762e-06
## white                  white 7.696411e-26   1.060734e-03
## claimant_rate claimant_rate 1.891786e-01   1.596702e-03
## MedianWage       MedianWage 1.844825e-33   1.349823e-02
## black                  black 7.413062e-18   1.925251e-01
## asian                  asian 2.417730e-16   3.725564e-01
```

Three things stand out:

Variables that stay strong. Educational attainment, age structure and Region remain highly significant in the full model, indicating genuine independent associations with Reform support that are not merely proxies for something else.

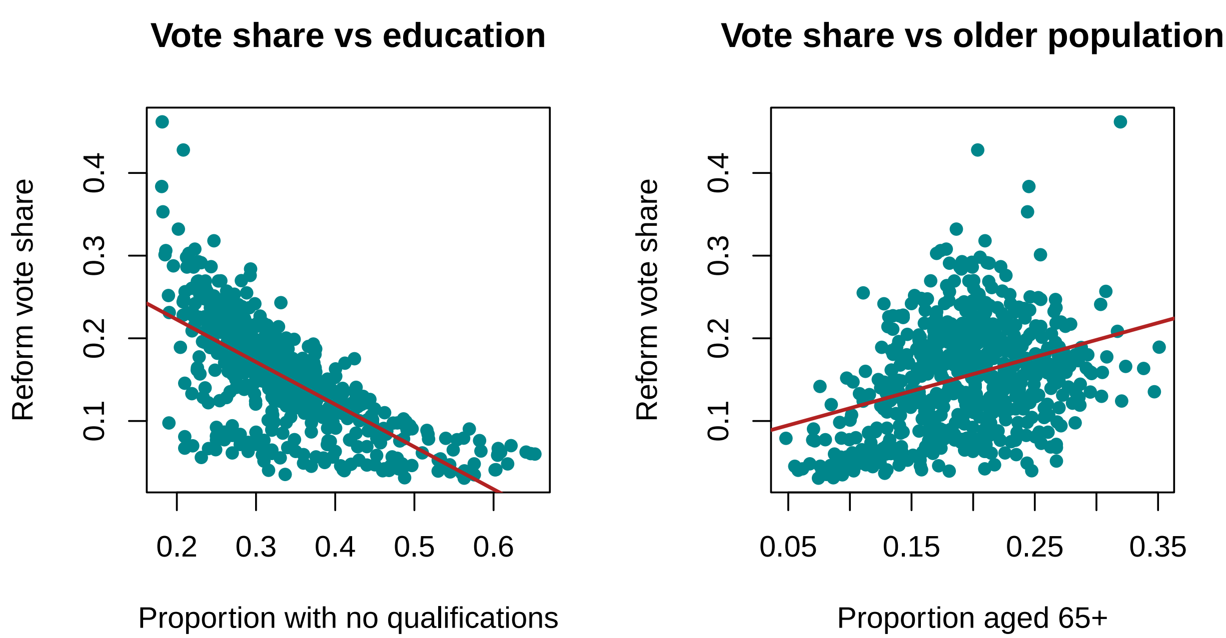
A suppressed effect that emerges. The benefit-claimant rate is weak or insignificant on its own but becomes clearly significant once education, age and region are held constant. This is a suppression effect: the raw association is masked by other variables, and only controlling for them reveals the underlying relationship between economic deprivation and vote share.

Associations that dissolve, and one that reverses. Some variables that looked important in isolation, such as ethnic composition, lose significance in the full model, implying their marginal associations were largely explained by correlation with education and region. Most strikingly, the older-population share flips sign: positively associated with Reform support on its own, but *negatively* associated once other variables are controlled for. Older populations tend to co-occur with other constituency characteristics (particularly around education and region) that themselves drive support, and once those are accounted for, the residual partial association with age runs the other way.

9 Interpretation

```
par(mfrow = c(1, 2))
plot(df$Qual5_pct, df$vote_share, pch = 16, col = "turquoise4",
     xlab = "Proportion with no qualifications", ylab = "Reform vote share",
     main = "Vote share vs education")
abline(lm(vote_share ~ Qual5_pct, data = df), col = "firebrick", lwd = 2)

plot(df$pct65.up, df$vote_share, pch = 16, col = "turquoise4",
     xlab = "Proportion aged 65+", ylab = "Reform vote share",
     main = "Vote share vs older population")
abline(lm(vote_share ~ pct65.up, data = df), col = "firebrick", lwd = 2)
```




```
par(mfrow = c(1, 1))
```

Educational composition is the single strongest structural predictor of Reform's constituency-level performance, and its effect is robust: it survives controlling for age, ethnicity, income and region. The age result is the clearest cautionary tale of the analysis. Viewed alone, older constituencies appear more supportive of Reform, but this bivariate impression is misleading; once the confounding influence of education and region is removed, the partial relationship reverses. It is a concrete demonstration of why simple pairwise relationships can mislead when explanatory variables are correlated, and why multiple regression, interpreted carefully, gives a more trustworthy picture of what independently drives an outcome.

10 Notes

Data: 2024 general election results and 2021 census demographics, sourced from the House of Commons Library (public). This project began as a piece of university coursework and has been rewritten as an independent, self-directed analysis. All processing is reproducible from the original public data files in R.